Google Stock Market Data Engineering Pipeline

Capstone Project — Data Analytics — Data Engineering Capstone

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1. Problem Statement

In modern financial markets, vast amounts of stock data are generated every trading day. Without a structured data pipeline, this raw data remains difficult to analyze, compare, or use for decision-making. Manual analysis is time-consuming, error-prone, and not scalable. There is a clear need for an automated, end-to-end pipeline that can ingest, clean, transform, store, and visualize stock market data efficiently.

This project focuses on Google (GOOGL) stock data — one of the most actively traded technology stocks — and aims to build a robust data engineering solution that transforms raw OHLCV (Open, High, Low, Close, Volume) data into actionable financial insights.

2. Solution & Features

The project implements a four-stage data pipeline entirely in Python, covering all core data engineering disciplines — ingestion, transformation, storage, and analysis.

- **Data Ingestion:** Fetches 2 years of GOOGL OHLCV data (503 trading days) and saves it as a structured CSV file.
- **Data Transformation:** Cleans raw data and engineers 5 analytical features: 7-day MA, 30-day MA, Daily Return %, 7-day Volatility, and Price Range.
- **Data Storage:** Persists the processed dataset into a SQLite relational database and exports summary statistics.
- **Analysis & Visualization:** Generates 5 publication-quality charts covering price trends, volume, returns, distribution, and volatility.
- **Automation:** A single main.py script orchestrates all pipeline stages end-to-end.

3. Screenshots & Output Charts

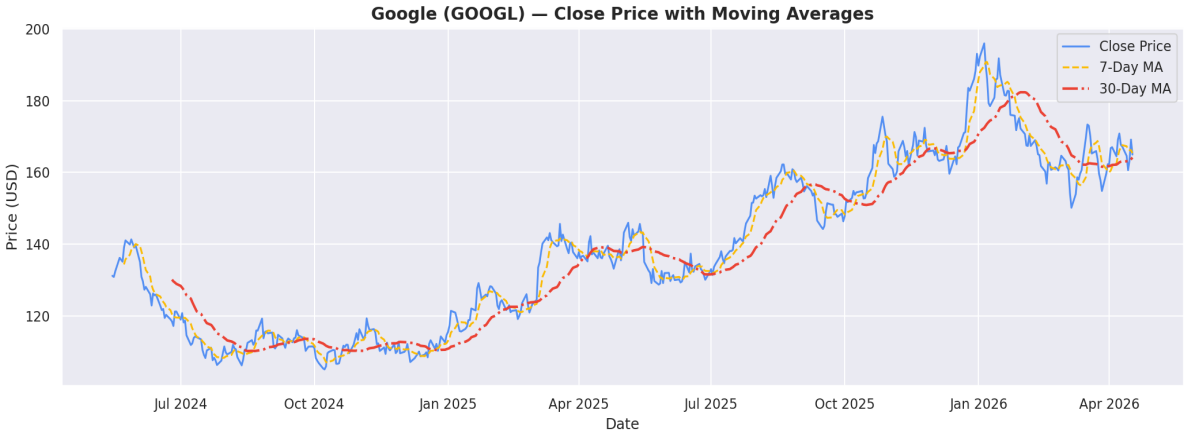

Figure 1: GOOGL Close Price with 7-Day and 30-Day Moving Averages

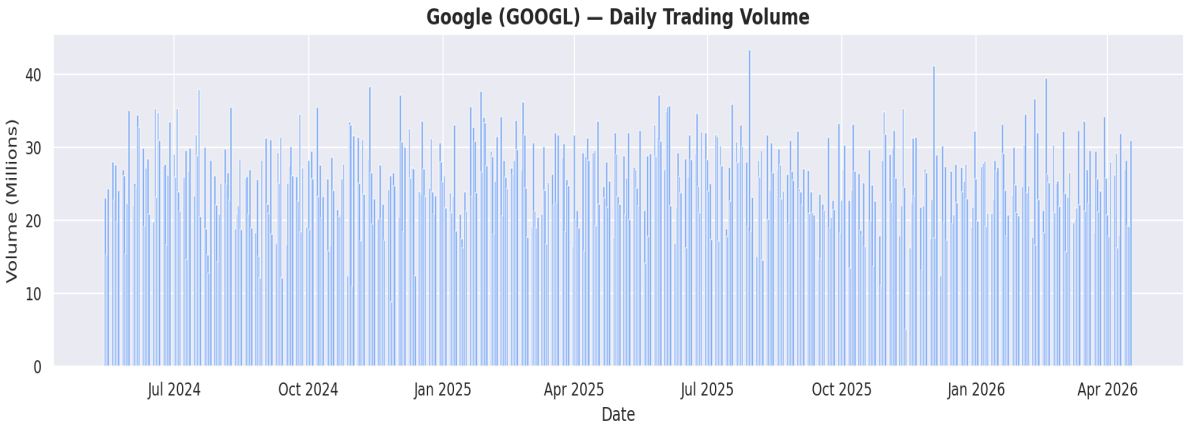

Figure 2: Daily Trading Volume (in Millions)

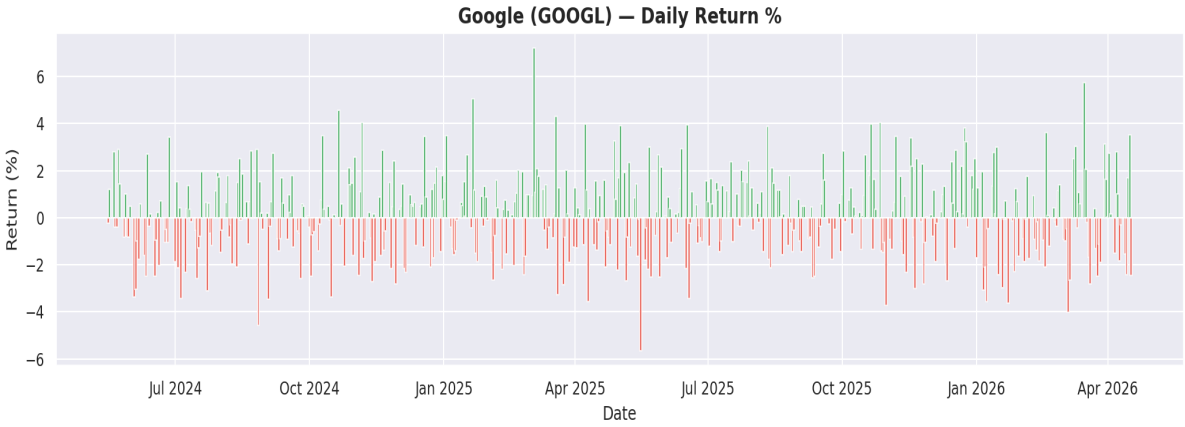

Figure 3: Daily Return % — Green = Gain, Red = Loss

3. Screenshots (continued)

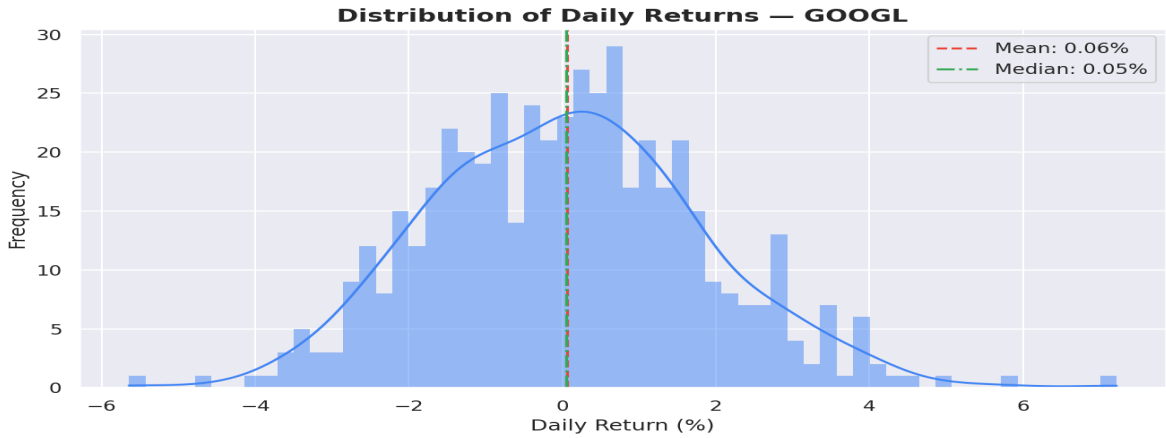

Figure 4: Distribution of Daily Returns with KDE Curve

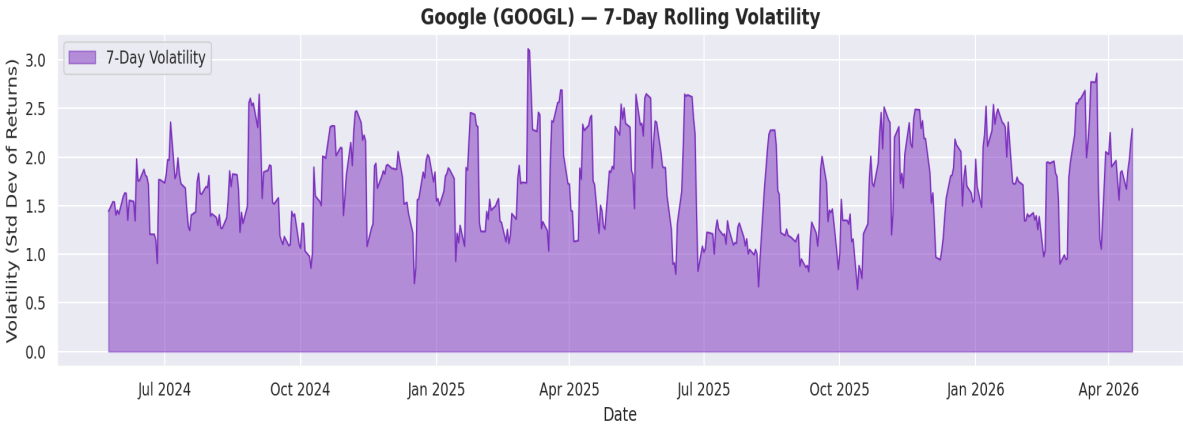

Figure 5: 7-Day Rolling Volatility Over Time

4. Tech Stack

Technology	Purpose	Version
Python 3.x	Core programming language	3.12
pandas	Data manipulation & transformation	2.x
NumPy	Numerical simulation & computation	1.x
yfinance	Stock data ingestion (API wrapper)	1.3.0
SQLite3	Relational database storage	Built-in
matplotlib	Chart generation & visualization	3.x
seaborn	Statistical visualization styling	0.13.x

5. Unique Points

- End-to-end pipeline in pure Python with zero cloud dependency — runs locally on any machine.
- Engineered 5 domain-specific financial features (moving averages, return %, volatility) beyond raw OHLCV data.
- Dual storage strategy: CSV files for portability and SQLite database for structured querying.
- 5 distinct chart types covering trend, volume, momentum, distribution, and risk — comprehensive market view.
- Modular design — each pipeline stage is an independent script, making the system easy to extend or swap components.
- Single-command execution via main.py — entire pipeline from raw data to charts in one run.

6. Future Improvements

- Live Data Integration: Connect to Yahoo Finance or Alpha Vantage APIs for real-time GOOGL data ingestion.
- Multi-Stock Support: Extend the pipeline to compare GOOGL against AAPL, MSFT, AMZN — sector analysis.
- Automated Scheduling: Use Apache Airflow or cron jobs to run the pipeline daily at market close.
- Machine Learning Layer: Build a price prediction model (LSTM or XGBoost) using the engineered features.
- Interactive Dashboard: Deploy a Streamlit or Dash web dashboard for live chart exploration.
- Cloud Storage: Replace SQLite with AWS S3 + Athena or Google BigQuery for scalable data warehousing.
- Alerting System: Email or Slack notifications when volatility exceeds a threshold or price drops sharply.

This project demonstrates a complete, production-style data engineering pipeline — from raw financial data to cleaned, stored, and visualized insights.